How Does AI Watermarking Work?

A Close Look at SynthID

Emanuele Fabbiani

Codemotion Conference · Milan, Italy, 2026

Detecting AI text is hard.

JANUARY 2023

OpenAI releases a detector for AI-written text

→

JULY 2023

Shut down for its low accuracy

OpenAI, New AI classifier for indicating AI-written text, 2023

ANTHROP\C

Aug 14, 2026 · Safeguards

How Claude's text watermark works

Future Claude models will generate text that contains a watermark.

OpenAI

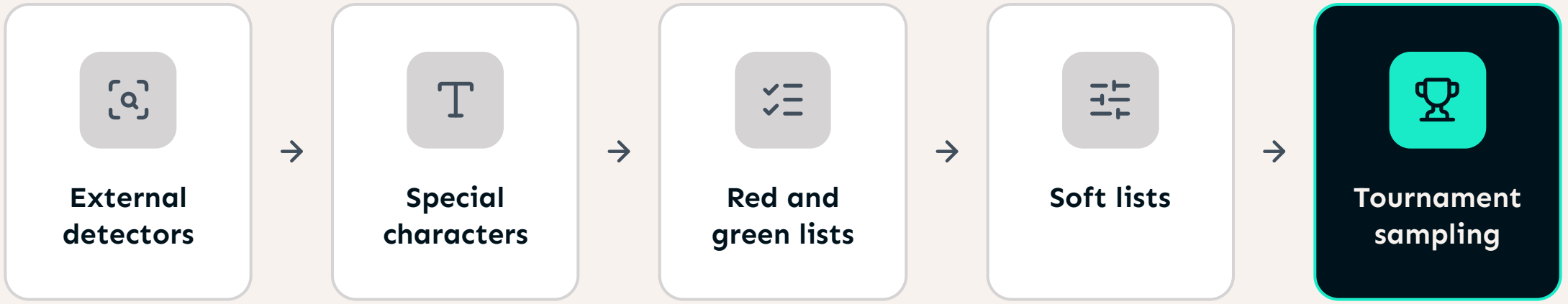
October 5, 2026 · Safety

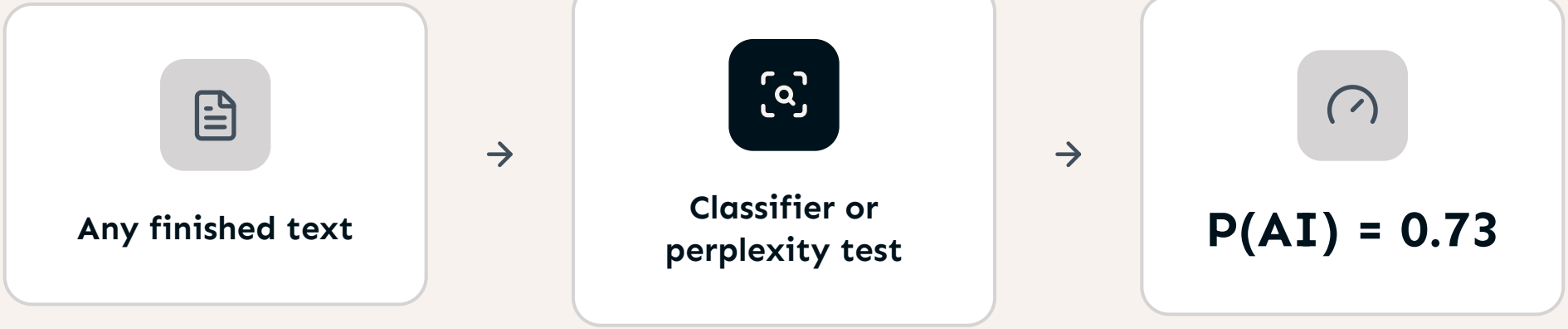
Our approach to EU text provenance rules

Today, we're sharing our approach to text watermarking in response to the EU AI Act.

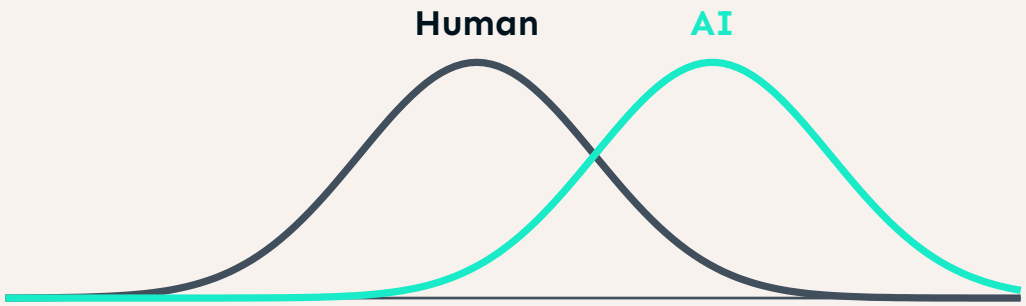
Anthropic, How Claude's text watermark works, 2026; OpenAI, Our approach to EU text provenance rules, 2026

How did we get there?

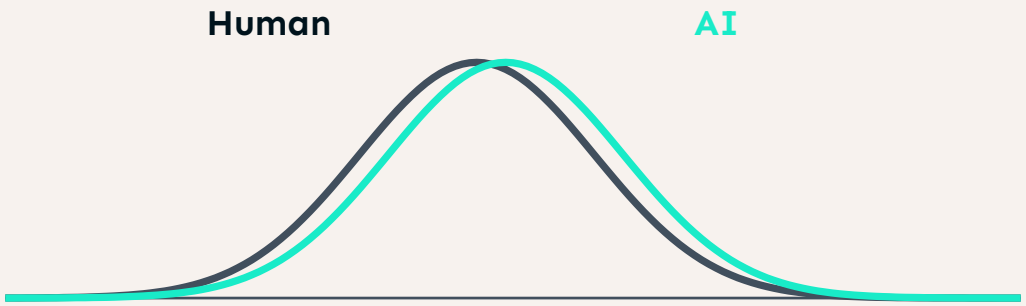

External detectors
Special characters
Red and green lists
Soft lists
Tournament sampling


Any finished text
Classifier or perplexity test
P(AI) = 0.73

WHEN MODELS WROTE BADLY



TODAY



Sadasivan et al., Can AI-Generated Text be Reliably Detected?, 2025

Lesson 1: put the signal in at generation time.

WHAT THE READER SEES

Watermarks hide in plain sight

WHAT THE BYTES SAY

Watermarks `U+200B` **hide** `U+2009` **in** `U+200B` **plain** `U+2009` **sight**

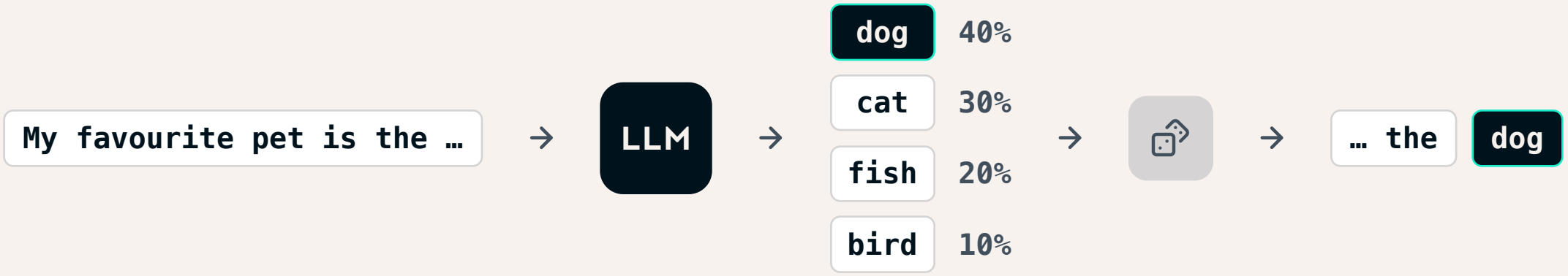
Sato et al., Embarrassingly Simple Text Watermarks, 2023

```
import re, unicodedata

text = unicodedata.normalize("NFKC", text)
text = re.sub(r"[\u200b-\u200f\u2060\ufeff]", "", text)
```

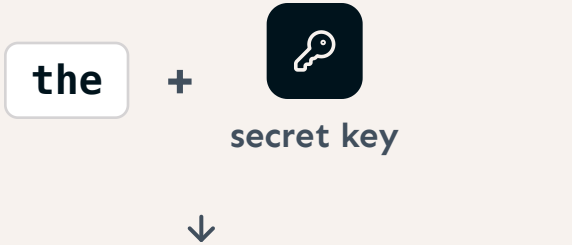
Lesson 2: hide the mark in the word choice.


My favourite pet is the …
LLM
dog
40%
cat
30%
fish
20%
bird
10%


My favourite pet is the …
LLM
dog
40%
cat
30%
fish
20%
bird
10%
… the
dog

1 · SEED

My favourite pet is **the**



Kirchenbauer et al., A Watermark for Large Language Models, 2023

1 · SEED

My favourite pet is **the**

the + 

secret key

↓

hash(last token, key) = 4821

→

2 · SPLIT THE VOCABULARY

green list

cat **bird** **car**

song **book** **plan**

red list

dog **fish** **idea**

river **city** **word**

Kirchenbauer et al., A Watermark for Large Language Models, 2023

1 · SEED

My favourite pet is **the**

the + 

secret key

↓

`hash(last token, key) = 4821`

→

2 · SPLIT THE VOCABULARY

green list

cat bird car
song book plan

red list

dog fish idea
river city word

→

3 · SAMPLE ONLY GREEN

~~dog~~ 40%
cat 30%
~~fish~~ 20%
bird 10%

↓

… the **cat**

Kirchenbauer et al., A Watermark for Large Language Models, 2023



same key, recolour every token

HUMAN TEXT

dogs are loyal and love long walks daily → 4/8 chance

WATERMARKED TEXT

cats are calm and purr on my lap → 7/8 watermark

Kirchenbauer et al., A Watermark for Large Language Models, 2023



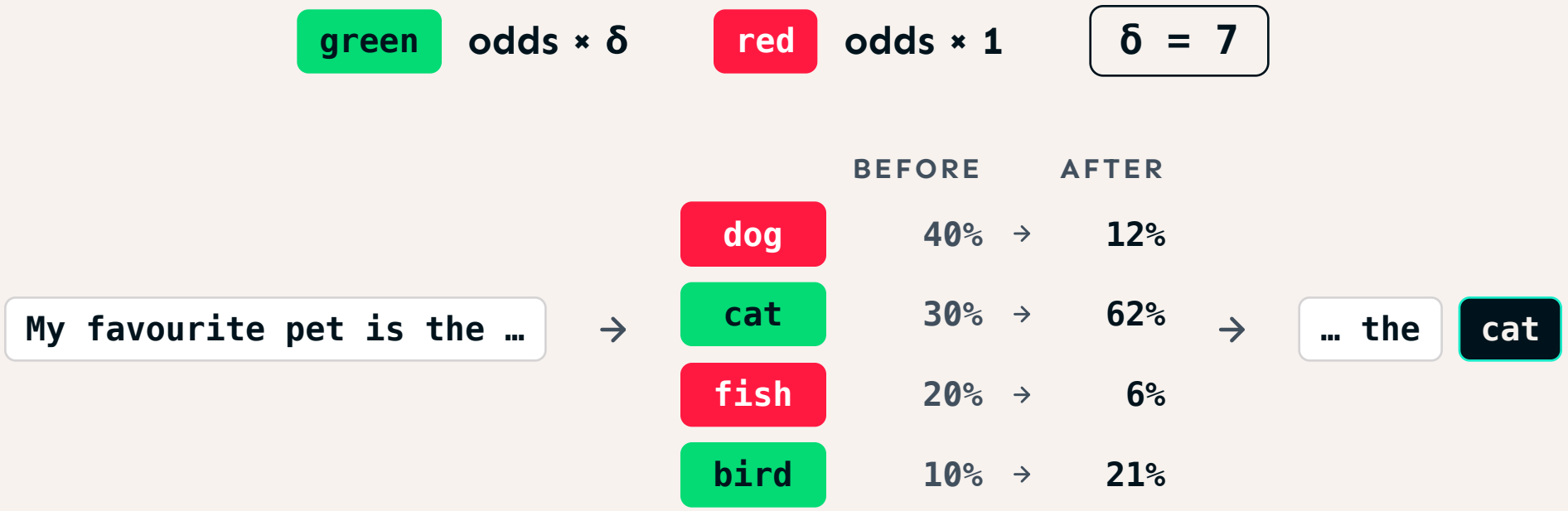
Kirchenbauer et al., A Watermark for Large Language Models, 2023

Lesson 3: step aside when the model is sure (low entropy).

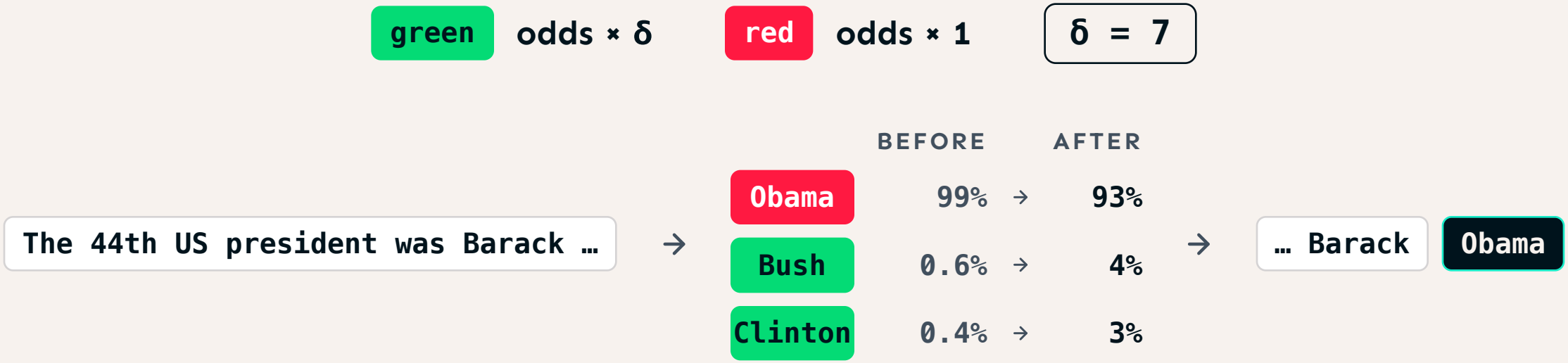


	ODDS		PROBABILITY
`dog`	**4**	→	4/10 = 40%
`cat`	**3**	→	3/10 = 30%
`fish`	**2**	→	2/10 = 20%
`bird`	**1**	→	1/10 = 10%

probability = odds ÷ **sum of all odds**



Kirchenbauer et al., A Watermark for Large Language Models, 2023



Kirchenbauer et al., A Watermark for Large Language Models, 2023

~800

tokens are enough to detect AI content
after strong human paraphrasing

Kirchenbauer et al., On the Reliability of Watermarks for Large Language Models, 2024

Lesson 4: to preserve quality, don't change the LLM's output distribution.

1 · SEED

My favourite pet is the

favourite **pet** **is** **the** **+** 🔑 secret key

↓

hash(last 4 tokens, key, round=1) = 4821

hash(last 4 tokens, key, round=2) = 1907

hash(last 4 tokens, key, round=3) = 6350

→

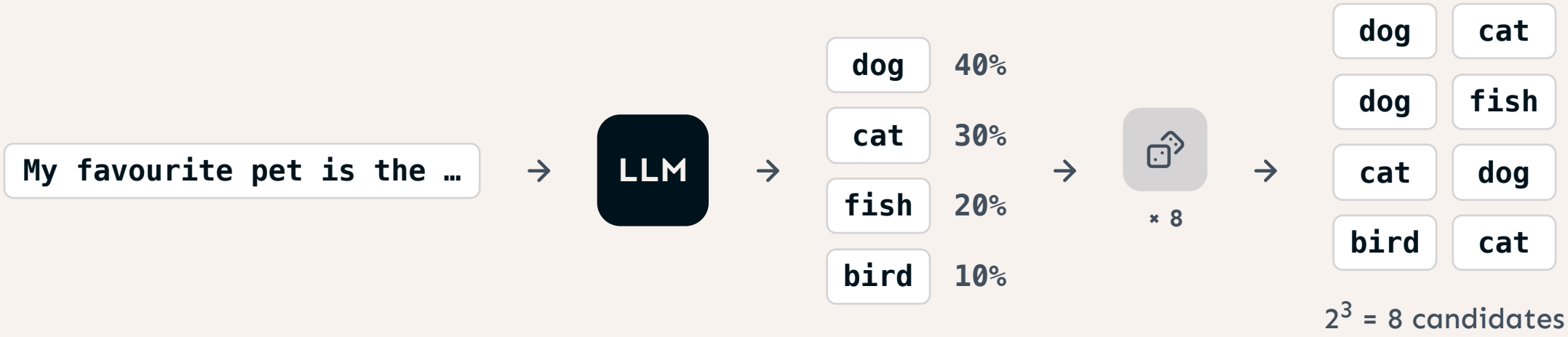
2 · SCORE EVERY CANDIDATE

	g_1	g_2	g_3
dog	0	1	0
cat	1	0	1
fish	1	1	0
bird	0	1	1

3 rounds here · 30 in SynthID

Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

3 · SAMPLE THE CANDIDATES



Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

4 · KNOCKOUT TOURNAMENT

8 CANDIDATES

- dog
- cat

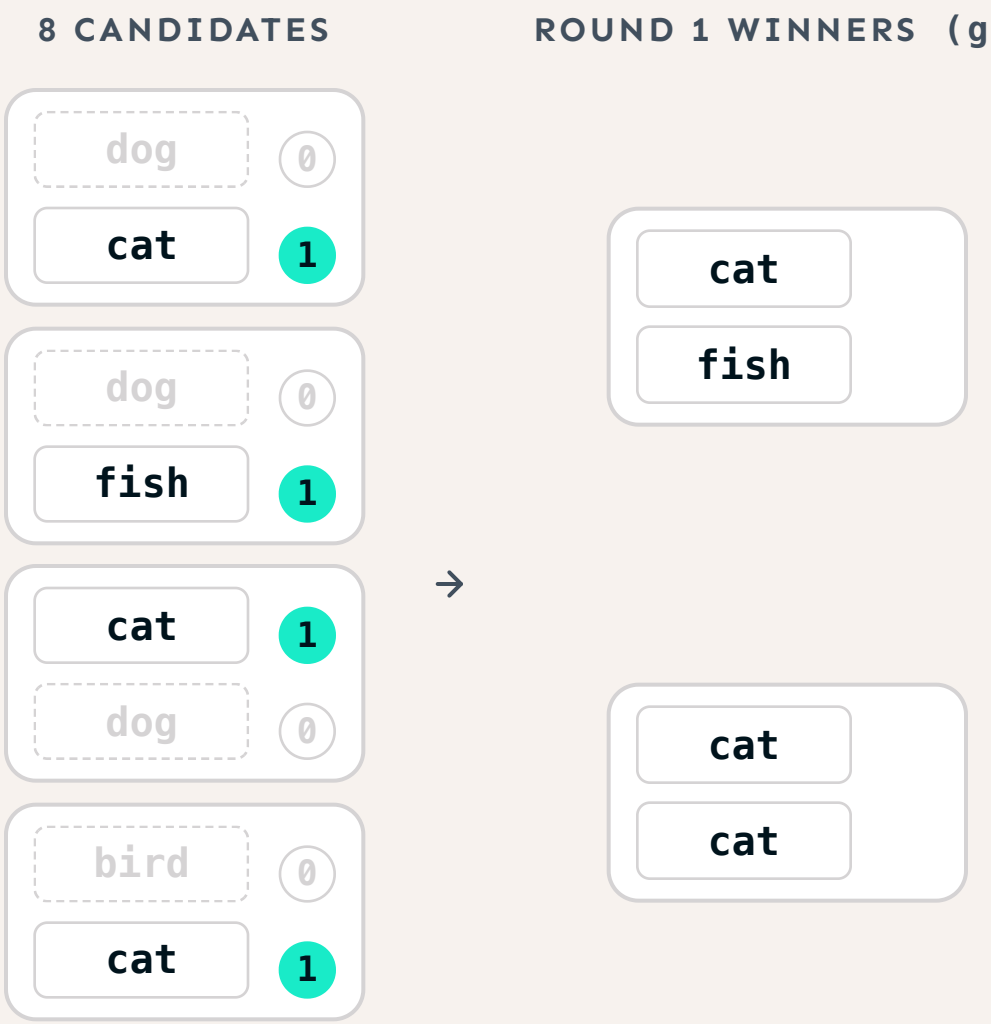
- dog
- fish

- cat
- dog

- bird
- cat

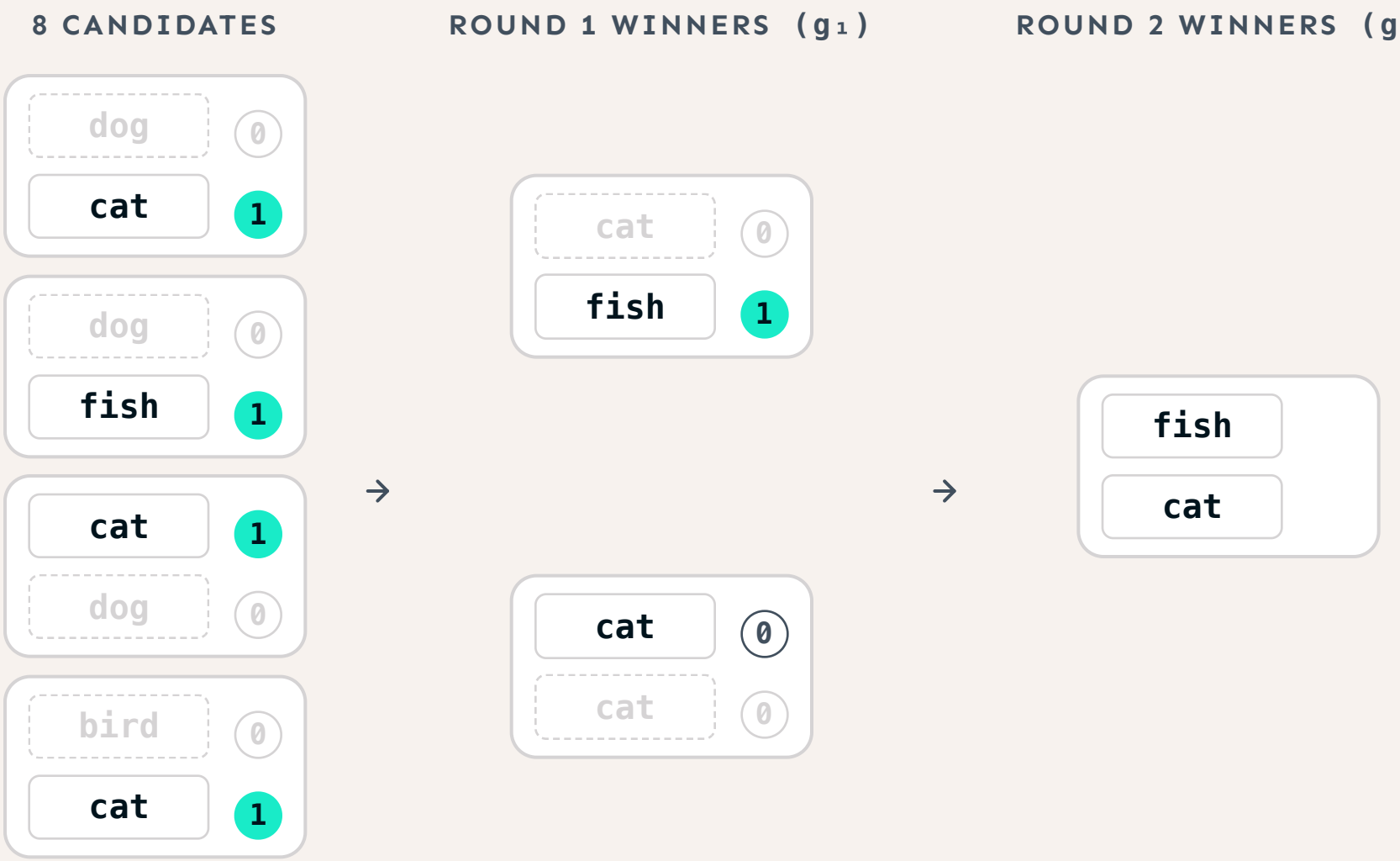
Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

4 · KNOCKOUT TOURNAMENT



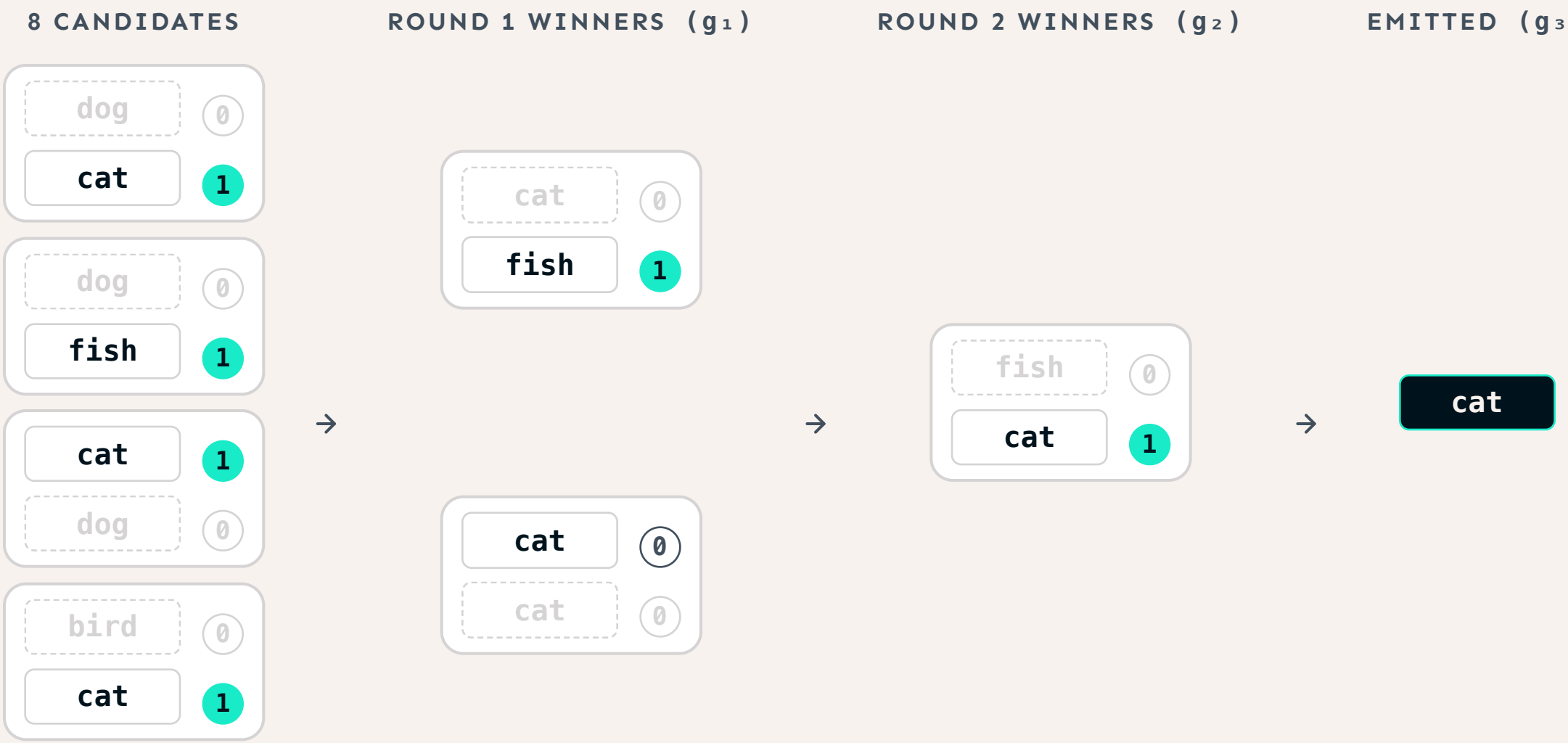
Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

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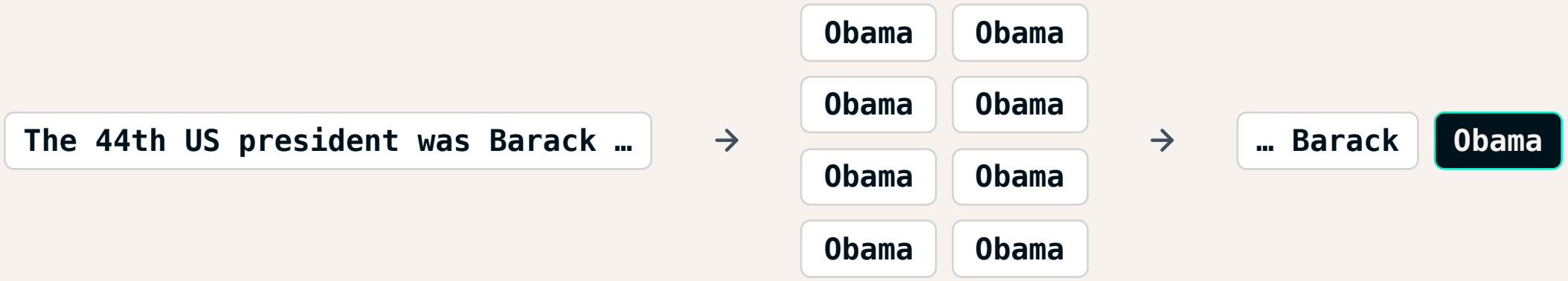


Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

4 · KNOCKOUT TOURNAMENT

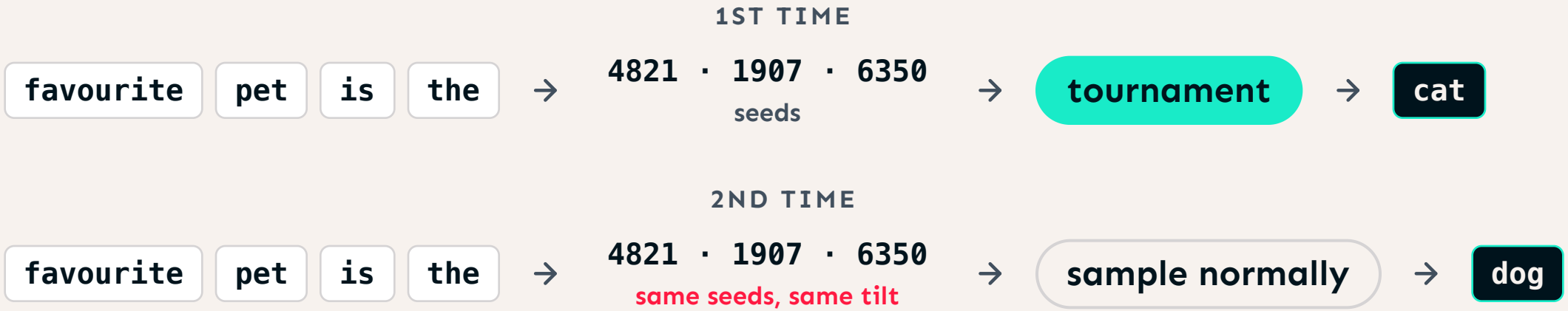


Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024



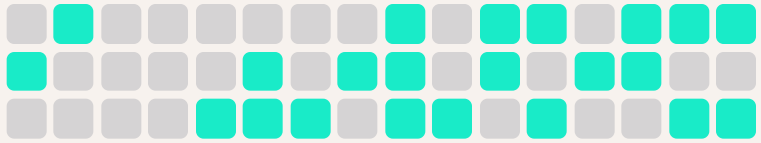
Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

My favourite pet is the cat, and your favourite pet is the …



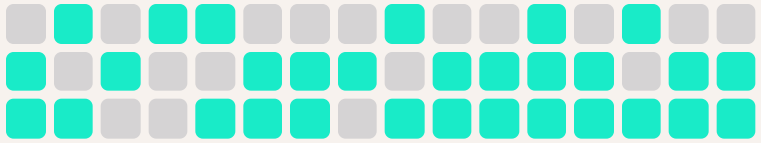
Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

HUMAN TEXT



mean = 0.46

WATERMARKED TEXT



mean = 0.63

Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

+0.57%

latency per token

15.527 → 15.615 ms on Gemma 7B

0.01%

thumbs-up gap

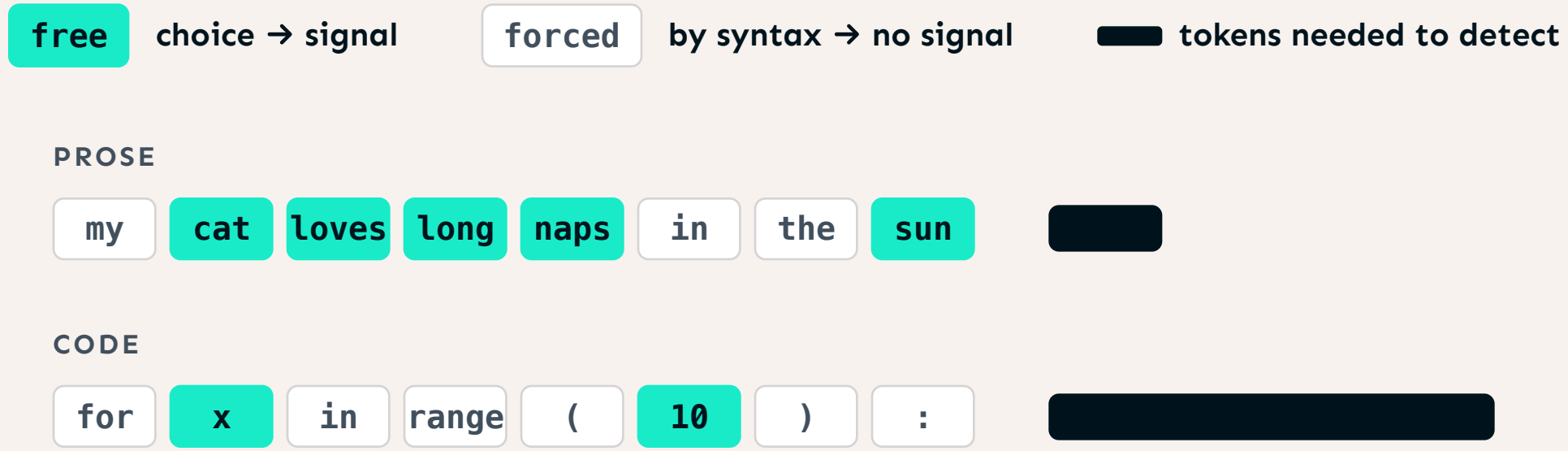
over ~20M Gemini responses

95%

true-positive rate

at 1% false positives, abstaining on uncertain texts

Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024



Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

A full rewrite or a translation erases it.

OpenAI, Understanding the source of what we see and hear online, 2024

Only the key holder can check for it.

Google, SynthID Detector — a new portal to help identify AI-generated content, 2025

ATTEMPT	CONSTRAINT IT TAUGHT US
External detectors	**Put the signal in at generation time**
Special characters	**Hide the mark in the word choice**
Red and green lists	**Step aside when the model is sure**
Soft lists	**Don't change the output distribution**

SynthID is clever engineering.
Learning how we got there
is the real reward.

ABOUT ME

Emanuele Fabbiani

Engineering Manager at TeamSystem



Founder at xtream and Reaidy, both acquired by TeamSystem in 2025.

Professor in AI at the Catholic University of Milan; guest lecturer at PoliMI, EPFL and Politechnika Wrocławska.

PhD in Applied AI, with 9 peer-reviewed papers and 170+ citations.

Speaker at 40+ international conferences, including AMLD Lausanne, ODSC London, WeAreDevelopers Berlin, PyData Berlin, Paris and Milan, PyCon Italy, Swiss Python Summit, Codemotion.

Connect on LinkedIn

References (1/2)

[1] Dathathri et al., Scalable watermarking for identifying large language model outputs, 2024

[2] Google DeepMind, SynthID Text (reference implementation), 2024

[3] Ghaisas et al., Introducing SynthID Text, 2024

[4] Google, SynthID: Tools for watermarking and detecting LLM-generated Text, 2024

[5] Google, SynthID Detector — a new portal to help identify AI-generated content, 2025

[6] Anthropic, How Claude's text watermark works, 2026

[7] Scott Aaronson, Anthropic's LLM watermarking, 2026

[8] OpenAI, New AI classifier for indicating AI-written text, 2023

[9] Liang et al., GPT detectors are biased against non-native English writers, 2023

[10] Mitchell et al., DetectGPT: Zero-Shot Machine-Generated Text Detection using Probability Curvature, 2023

[11] Krishna et al., Paraphrasing evades detectors of AI-generated text, but retrieval is an effective defense, 2023

[12] Sadasivan et al., Can AI-Generated Text be Reliably Detected?, 2025

[13] NIST, Reducing Risks Posed by Synthetic Content (NIST AI 100-4), 2024

References (2/2)

[14] C2PA, C2PA Technical Specification, 2026

[15] Sato et al., Embarrassingly Simple Text Watermarks, 2023

[16] Zhao et al., Invisible Image Watermarks Are Provably Removable Using Generative AI, 2024

[17] Kirchenbauer et al., A Watermark for Large Language Models, 2023

[18] Kirchenbauer et al., On the Reliability of Watermarks for Large Language Models, 2024

[19] Kuditipudi et al., Robust Distortion-free Watermarks for Language Models, 2024

[20] Christ, Gunn & Zamir, Undetectable Watermarks for Language Models, 2024

[21] Zhang et al., Watermarks in the Sand: Impossibility of Strong Watermarking for Generative Models, 2024

[22] Jovanović, Staab & Vechev, Watermark Stealing in Large Language Models, 2024

[23] Omidi, Dong & Wang, On Google's SynthID-Text LLM Watermarking System, 2026

[24] OpenAI, Understanding the source of what we see and hear online, 2024

[25] European Parliament & Council, Regulation (EU) 2024/1689 (AI Act), Article 50, 2024

[26] Margherita Bassi (Smithsonian), A.I. May Have Solved a Longstanding Math Problem With a Million-Dollar Prize, 2026

[27] OpenAI, Our approach to EU text provenance rules, 2026

[28] nohypeai, Watermarking Playground, 2026